

ARPIT RAI
PWAY DESIGNER



Innovative – Detail-oriented – Proactive – Reliable

CORE SKILLS

1. MicroStation
2. Power Rail Track
3. ClearRoute2
4. D-Gauge
5. OpenRail Designer
6. Python
7. VBA
8. C#
9. Generative Components

ROLE ON PROJECTS

- P-Way Designer

POSITION

- Graduate Engineer (Inactive)
- Assistant Engineer (Active)

QUALIFICATIONS

- B.E. (Civil Engineering)
Batch 2018-2021

YEARS OF EXPERIENCE

- 3 (ARCADIS)

AWARDS

- Digital Champion (Q4-2022)
- Spot Award (Q4-2023)

ACHIEVEMENTS

- Expedition DNA Ambassador
(Aug 2023 Onwards)
- Nominated for Best Employee
L-6A (Q1-2023)
- Nominated for Best Employee
L-5 (Q1-2022)

- I have practical experience in the railway industry, specifically in alignment design for conventional railways. For my projects, I primarily use Bentley Power Rail Track, MicroStation, D-Gauge, ProjectWise, and OpenRail for design and drawing production.

Suitability to the Role

With nearly three years of experience as a track designer, I excel in creating, reviewing, and comparing designs - key skills for a PWAY engineer in the rail industry. I am also proficient with automation tools, having streamlined processes to boost productivity. My cross-cultural experience in the UK and the Netherlands highlights my adaptability. I am ready to leverage these skills as a PWAY designer to ensure efficient, quality-focused results.

Relevance of Experience

Applications like MicroStation, ClearRoute2, D-Gauge, and Bentley Power Rail Track are highly sought-after especially in the UK Rail sector. Possessing skills in these applications enables accurate decision-making and expedites project delivery while upholding quality standards.

On the automation front, I have effectively utilized Python, C#, VBA, Power Platform, and AutoLISP to develop design automations for projects in the Netherlands and the UK over the past two years.

Project Experience

Victoria Road Underbridge Track Renewals

TfNSW (May 2024 – May 2023) AUS

The scope of work was to handle the drawing production for the ongoing Victoria Road Underbridge track renewal project in the absence of the assigned resource and to submit the deliverables on time.

- I was responsible for the Stage 3 submission of drawings as per the schedule.
- This was a good opportunity to refresh my AutoCAD skills and become familiar with drawing production in an AutoCAD environment.

Project Experience Contd.

Beckton MFB – Construction Support

Network Rail (November 2023 – December 2023) UK

The scope of work was to assist DLR in the construction of Beckton MFB, including responding to RFIs raised by DLR or other involved parties during the construction phase.

- I was involved in answering RFIs raised by DLR or other involved parties during the construction phase from Arcadis' Track discipline this may involve defending the design decisions taken in GRIP 5 phase, rerouting the RFI response to correct document number or making minor design changes as per the requirements.

Exeter St. David's Platform Extension

Network Rail (March 2023 – Dec 2023) UK

The scope of works was to propose a platform extension scheme to the Bay area of Exeter St. David's Platform 2 to accommodate the 5 car IETs. The project currently stands at ES 4&5 stage where detailed design has been carried out.

- I was responsible for creating a viable alignment geometry through the Bay platform and extend it to achieve the required extension with Buffer stop risk assessments and ultimately produce the complete Form A/B pack for the detailed design phase.

Bugley Bridge Embankment Renewals

Network Rail (March 2023 – March 2023) UK

The scope of work was to recreate the previous Form B alignment and undertake the full Form B and AFC submission after addressing the DRN comments.

- I was responsible for drawing production of the Form B detailed design.
- I was in part, with the CRE responsible for the Form B report and AFC drawing for Bugley.

River Avon Bridge Renewals

Network Rail (February 2023 – March 2023) UK

The scope of work consisted of checking the design done in the previous Form B stage and confirming whether the previous Form B design is still applicable and if so, producing AFC drawings for River Avon Bridge renewal.

- I was responsible for checking the previous Form B design.
- I was responsible for the Form B report and AFC drawings for River Avon. (Abandoned)

Whitechapel Station

Transport for London (March 2022 - January 2023) UK

The scope of work consisted of checking if the design on the station has been done as per requirements and it meets the necessary lifts for the lines through Whitechapel station.

- I was responsible for making sure the As-Built survey is correct and then make sure adequate lifts have been provided and do final Platform clearance runs through the Platform area.
- I was responsible for carrying out the stepping assessment of the platforms as per RVAR compliance.
- I was responsible for the As-built comparison report and creation of proving alignment to make sure the As-Built works were done as per the design provided.

Project Experience Contd.

Woking Victoria Arch Bridge

Network Rail (September 2022-December 2022) UK

The scope of work consisted of producing a detailed design for Woking Victoria Arch Bridge renewal which required the portion of tracks within the renewal extents to be removed and relaid with new track and crossing components upon completion of works. There were also proposals for completely dismantling old sidings to make room for new sidings.

- I was involved in detailed design of Horizontal and vertical alignment for all the main lines including S&Cs.
- I was responsible for complete gauging assessment throughout the design phase.
- I was involved in drawing production for the detailed design.

Beaulieu Park Station

Network Rail (June 2022-September 2022) UK

The scope of work was to produce existing and proposed gauging reports for the new station being constructed at Beaulieu site.

- I was responsible for producing the existing gauging and proposed gauging for structure clearance, passing clearance, proposed platform clearance and Signal clearance.

Beckton Depot

Docklands Light Railway (February 2022-March 2022) UK

Beckton MFD was in a GRIP 6 stage where an OTC was required through track to get the materials on the Depot site.

- I was responsible to create the 3D model and mark the OTC location.
- I was responsible for the panelling work through the Wheel lathe building with CRE's assistance.

CP6 Wessex Route (SMDF)

Network Rail (June 2022 - June 2022) UK

The Southern Multidisciplinary Framework (SMDF) comprised of a range of projects through all stages of the project lifecycle to be delivered on the Wessex route.

- I was responsible for making sure the As-Built survey received is correct and then produce a gauging report as per As-Built data to make sure the As-Built clearances are under compliant values.

Central Scotland Gauging 7 Sites

Network Rail (September 2021- February 2022) UK

On the site some O/Bs were then presenting gauging issues (usually because of an arch) which was typically being removed and replaced with a flat deck. The new bridge decks needed to be set to a position that allowed for a future track renewal, so, we were asked to carry out a compliant track alignment beneath each bridge.

- I was responsible for producing GRIP Stage 4 track design and drawings.
- I was responsible for producing design tables for Lifts/Slues, Platform tables (wherever applicable) and full Form A document pack.

Crossrail GRIP 6 to 8 Support

Network Rail (September 2021-December 2021) UK

The scope of work was to update designed drawings to As-Built stage drawings based on Red-line markup received from contractor.

- I was responsible for updating the DGN drawings as per contractor's Red-line markup for category 1 and 2 drawings.

Human + Innovation Experience

PCRA Tool – Python + Excel

All Engineering Disciplines – GEC India (Jan 2024 – May 2024)

Production – Check – Review – Approval.

This tool allows a correct PCRA process to be followed which also ensures that no comments from client remain unaddressed and the deliverables produced are of the highest quality.

GIL Format Add-in – Excel

Cable & Ducts – Netherlands (Dec 2023 – Jan 2024)

The Add-in features buttons that can perform formatting and modification operations on the Cable Route Table according to ProRail standards. It eliminates the visually jarring process of modifying and formatting these large tables.

Cable Drawing Revision Add-in – MicroStation

Cable & Ducts – Netherlands (Oct 2023 – Jan 2024)

This ProRail-compatible Add-in offers options to create a Fictitious Cable Revision and a Red Revision drawing that complies with ProRail standards.

Curved S&C Annotation Tool – Generative Components

Rail PWAY - UK (Oct 2023)

This tool takes the Mainline and Branchline rails as input and creates a fully annotated S&C layout along with sleepers according to REPW drawings for the given type of turnout. It saves a lot of time, especially in yard areas where working on such locations can be tedious.

LoGD Tool – Revit + Excel

Highways Drainage - UK (May 2023)

This tool was developed in C# for Revit and utilizes input from Micro Drainage in the form of an Excel sheet. It accurately places drainage features such as manholes, drainage gullies, etc., according to the input, with custom dimensions, ensuring correct placement. The file would then be exported for use in Civil3D, where it can be further integrated with other interfaces for design purposes.

NESA PDF Extraction Tool – Python + Excel

Rail PWAY - UK (December 2022)

The National Electronic Sectional Appendix is utilized for gauging purposes to list the trains running on a particular ELR. These are large PDFs where it can be easy to miss a vehicle. This tool carefully scans the PDF after taking certain inputs from the designer and provides a list of vehicles. This process saves one hour for each run.

UK Design Check Tool – Python + Excel

Rail PWAY - UK (Oct 2022)

Automated design checks are essential, particularly when dealing with multiple lines or variations in a design. This tool meticulously examines a design produced in Bentley Rail Track for specific design parameters spanning across horizontal, vertical, and cant design to ensure design integrity before it is delivered to the client.

Contact Wire Wind Deviation Tool – LISP + Excel

Rail OLE - Netherlands (July 2022)

Designing a contact wire wind deviation envelope was both time-consuming and error prone. To address this challenge, I developed a LISP program that utilizes an existing Excel sheet to check the design parameters and automatically create the wind envelopes around the contact wire. Running this script for just 2 seconds would save around 2 days of work for the designer.

Human + Innovation Experience contd.

QMS Automation Tool – Python + Excel

All Engineering Domains - UK (Feb 2022)

Designers and project managers must ensure that important files are stored in the correct location for the Quality Management System. Assigning responsibility for this task can be challenging, particularly if resources change over time. This can lead to difficulties during audits and cause tension for new team members. This automation helps streamline the process: with a click, you can identify which documents are present for your project, where to find them, and which ones are missing and where to place them.